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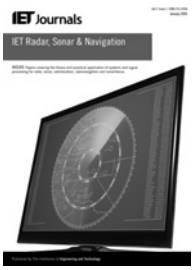

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Efficient direct position determination of orthogonal frequency division multiplexing signals

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Abstract: The authors present an efficient method for positioning of orthogonal frequency division multiplexing (OFDM) signals. The method, called direct position determination (DPD), enables one to determine the emitter position in a single search operation. In formerly published articles on DPD, the signal was assumed to be a narrowband Gaussian source, which was subjected to a rather simple flat-fading channel model. Here, the authors extend the DPD algorithm to handle the OFDM signals that propagate through a frequency-selective channel, and derive the least squares estimator (LSE) for the problem. In addition, the authors present a method for a fine 2D position estimation, which enables one to reduce the DPD computational complexity without compromising on its estimation accuracy.

1 Introduction

The growing demand for wireless location-based services, currently attracts much attention to accurate mobile location systems. An indication of the mutual academic and industrial interest in this topic can be found in the galore of literature that has recently been published on this subject. Examples of these recent publications, which practically cover all the wireless standards being developed to date, include: location in cellular networks [1–4], sensor networks (see e.g. [5] and its references), ultra wideband (UWB) networks [6], and many more.

Despite the undisputable fact that the global positioning system (GPS) is currently the wireless industry's preferred choice for mobile-device location, the bill of material costs and the wireless-coexistence challenges involved in the integration of GPS receivers into small form-factor, mobile handsets containing multiple wireless subsystems motivate the development of alternative, accurate emitter localisation systems. Parallel to the GPS trend, another prominent direction of the wireless industry is its clear preference for orthogonal frequency division multiplexing (OFDM)-based solutions, which to date are being widely adopted by a variety of wireless standards including: IEEE 802.11a/g and 802.11n (Wireless LAN (WLAN)/Wi-Fi), IEEE 802.16e (WiMAX), WiMedia UWB, digital video broadcasting systems, Release-8 of 3 GPP (a.k.a. long-term evolution) and others.

Converse to the trends and challenges aforementioned, the problem of an accurate and efficient OFDM emitter location has not been thoroughly explored yet. Location of the OFDM signals has been partially addressed in [7, 8]. In both cases, the positioning was based on the time-of-arrival (TOA) information extracted from the channel estimation, with no exploitation of the angle-of-arrival (AOA) information.

Direct position determination (DPD) has recently been proposed as a highly accurate technique for emitter localisation [9]. As opposed to the traditional two-step AOA/TOA location methods (see e.g., [2] and its references), DPD uses a single-step location algorithm that estimates the location directly from the signals captured at several antenna arrays simultaneously. Thus, DPD eliminates the need for a triangulation step that is required by the traditional positioning methods for determining the emitter location. In this paper, we extend the seminal

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DPD estimator, which was developed for narrowband signals that are subjected to flat-fading channels, to handle the OFDM signals. Similar to OFDM, DPD also operates on the received signal in the frequency domain. This makes DPD an attractive algorithm for locating the OFDM signals.

The channel model that will be described in the sequel consists of multiple clusters of scatterers along with a 'line-of-sight' (LOS) cluster of scatterers that surround the mobile emitter to be located. In our model, it assume that some of the paths emanate from the LOS cluster, whereas the remaining paths generate non-line-of-sight (NLOS) interference that dissipates portion of the emitter signal energy. The sum of multiple delayed faded paths in the propagation channel generates different complex attenuations in different frequencies, or frequency selectivities. The scenario where only an NLOS channel component exists is often the case for indoor channel scenarios. In such scenarios, emitter positioning requires pre-calibration of the channel model in addition to the usage of empirical techniques such as pattern matching [10] or ray tracing [11].

To obtain an attractive and tractable algorithm, we make no assumptions regarding the location of the scatterer clusters, thereby eliminating the need for channel model calibration. The NLOS clusters' reflections are absorbed into a unified noise plus NLOS interference term and the least-squares location estimation is carried out based solely on the LOS information.

The purpose of this paper is twofold. First, it outlines the least squares estimator (LSE) for the problem of direct positioning of the OFDM signals. Secondly, it addresses the issue of fine 2D parameter estimation. As will be discussed in Section 5, this topic has not received much attention in the positioning-related literature and is highly applicable for DPD problems.

1.1 Paper organisation

In Section 2, we formulate the signal and channel model assumed throughout this paper. Section 3 outlines the main points of the DPD algorithm and derives the LSE algorithm for OFDM signal positioning. Further implications of the DPD algorithm are discussed in Section 4. Next, in Section 5, we discuss several options for obtaining a fine parameter estimation in 2-D parameter estimation problems and detail the application of the, so-called, paraboloid interpolation to the DPD LSE. Numerical examples that highlight the superiority of the DPD method over the traditional AOA + TOA technique and also compare the accuracy of these methods are presented in Section 6. Finally, the summary and concluding remarks are outlined in Section 7.

Notation: In addition, we use the following conventions throughout the paper.

$\{\cdot\}^*$	conjugate
$\{\cdot\}^{\mathrm{T}}$	transpose
$\{\cdot\}^{\mathrm{H}}$	conjugate transpose
$\boldsymbol{A} \otimes \boldsymbol{B}$	Kronecker product of the matrices $\boldsymbol{A}$ and $\boldsymbol{B}$
$\mathrm{diag}\{\boldsymbol{x}\}$	$N \times N$ diagonal matrix whose diagonal is the vector $\boldsymbol{x}$
$\boldsymbol{I}_N$	$N \times N$ identity matrix
$\boldsymbol{X}^+$	the pseudo-inverse matrix of $\boldsymbol{X}$ that is $\{\boldsymbol{X}^{\mathrm{H}}\boldsymbol{X}\}^{-1}\boldsymbol{X}^{\mathrm{H}}$
$\|\cdot\|^2$	Frobenius norm of the matrix $\boldsymbol{A}$ that is, $\sum_{ij}\|[\boldsymbol{A}]_{ij}\|^2$

2 Problem formulation

Consider a single mobile subscriber unit (MSU) that transmits OFDM-modulated signal of bandwidth B through its single antenna. Each OFDM symbol consists of N_{s} subcarriers out of which a pre-defined number of subcarriers are pilot tones. The pilot tones are modulated with data that is known a-priori to the receiver and can be used for channel estimation. The density of the pilot tones in the OFDM symbol may vary according to the changing channel conditions [12]. The overall OFDM symbol length in time is $T_{\mathrm{s}} = T + T_{\mathrm{G}}$ where T is the fast fourier transform (FFT) integration time and T_{G} the length of the guard interval. The corresponding OFDM symbol duration in time samples is $N_{\mathrm{s}} + L_{\mathrm{CP}}$, where L_{CP} is the number of time samples in the cyclic prefix (CP) (which is equivalent to the length of the guard interval).

The transmitted OFDM signal is intercepted by J base transceiver stations (BTS). Each BTS is equipped with an antenna array consisting of M elements. It is assumed that all base stations are time-synchronised to the level provided by the GPS (~50 ns). The source, which is assumed to be stationary during the observation time, is also assumed to be located in the far field of the intercepting arrays (i.e. its distance is many times the arrays' apertures). The problem of interest is to accurately estimate the MSU position $\boldsymbol{p} = [x, y]^{\mathrm{T}}$ given a single observation of an OFDM symbol that is received at each of the J BTS arrays.

2.1 Channel

We model the delay spread by assuming that there are $L \leq L_{\mathrm{CP}}$ significant scatterer clusters in the propagation channel. This assumption guarantees that the frequency-selective fading channel indeed decouples into a set of frequency-flat fading subchannels. The locations of the L scatterer clusters (including the LOS cluster that contains the MSU), are unknown.

This channel model implies that the OFDM signal received at each of the BTS arrays undergoes a block-selective fading channel in the frequency domain. Block selectivity means that across the system bandwidth there exist several coherence bandwidths of B_{C} MHz. For the sake of simplicity and algorithm brevity, we further make

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the assumption of a 'piecewise-constant' fading in the frequency domain [13]. Under this assumption, all the subcarriers within the same coherence bandwidth are subjected to the same complex attenuation. This kind of assumption is commonly made for the design of space-frequency codes with multiple-input-multiple–output OFDM systems [14–17]. Again, the piecewise-constant fading assumption is only made for brevity and does not prevent standard channel estimation extrapolation techniques [18] from being applied. We further follow the common assumption that the number of coherence bandwidths along an OFDM symbol is proportional to the number of scattering clusters that exist in the channel [15].

2.2 Received OFDM signal model

Our signal model is based on the model devised by Gezici *et al.* [7] with the addition of AOA information. A single OFDM symbol with N_s-subcarriers and data vector

$$\bar{\mathbf{s}} = [\bar{s}(0), \bar{s}(1), \ldots, \bar{s}(N_s - 1)] \tag{1}$$

can be expressed as

$$s(t) = \frac{1}{\sqrt{T}} \sum_{k=0}^{N_s-1} \bar{s}(k) e^{j(2\pi/N_s T)kt}, \qquad -T_G < t < T \tag{2}$$

The signal received at the jth BTS propagates through a multipath channel with an impulse response,

$$\boldsymbol{h}_j(t) = \sum_{l=0}^{L-1} \boldsymbol{a}_j(\boldsymbol{p}_l)\alpha_j(l)\delta(t - \tau_j(\boldsymbol{p}_l)) \tag{3}$$

where $\boldsymbol{a}_j(\boldsymbol{p}_l)$ is the $M \times 1$ jth BTS array response vector to the signal arriving from position $\boldsymbol{p}_l = [x_l, y_l]^{\mathrm{T}}$, $\alpha_j(l)$ the complex path attenuation, t the time and $\delta(t)$ the Dirac delta function. In addition, we denote the nominal propagation delay from the lth cluster located at position $\boldsymbol{p}_l$ to the jth BTS array by $\tau_j(\boldsymbol{p}_l)$. We assume that the first $\mathcal{L}$ paths are reflected from the nearby scatterers surrounding the MSU that is located at position $\mathbf{p}$. The remaining $L - \mathcal{L}$ paths are multipath interference. We absorb the remaining multipath interference into the vector $\tilde{\boldsymbol{n}}_j(k)$, which is an $M \times 1$ complex vector defined to be the interference plus additive white noise.

Assuming that the observation time equals a period of a single OFDM symbol, the time-dependent signal received at the jth BTS array is given by (4), as shown at the bottom of the page.

Notice that in (4), we have used the common approximation [19] and [20], $\boldsymbol{a}_j(\boldsymbol{p}_\ell) \approx \boldsymbol{a}_j(\boldsymbol{p})$.

In the frequency domain, the kth element of the output vector after the receiver sampling and FFT processing is given by

$$\bar{\boldsymbol{r}}_j(k) = \boldsymbol{a}_j(\boldsymbol{p})\bar{s}(k)\left(\sum_{\ell=0}^{\mathcal{L}-1} \alpha_j(\ell) e^{-j(2\pi/N_s T)k\tau_j(\boldsymbol{p}_\ell)}\right) + \tilde{\boldsymbol{n}}_j(k) \tag{5}$$

where $\bar{\boldsymbol{r}}_j(k)$ is a frequency-dependent, $M \times 1$ vector.

We denote the nominal propagation delay from the MSU cluster located at position $\boldsymbol{p}$ to the jth BTS array by $\tau_j(\boldsymbol{p})$. Similarly, we use $\bar{\tau}_j(\boldsymbol{p}_\ell)$ to express the excess delay (over the nominal propagation delay) resulting from the ℓth scatterer surrounding the MSU. Therefore, we can rewrite (5) as shown in (6).

Finally, we can express the signal observed by the jth BTS array as

$$\bar{\boldsymbol{r}}_j(k) = \boldsymbol{a}_j(\boldsymbol{p})\bar{b}_j(k)\mathrm{e}^{-j\omega_k \tau_j(\boldsymbol{p})}\bar{s}(k) + \tilde{\boldsymbol{n}}_j(k) \tag{7}$$

where

$$\omega_k \triangleq \frac{2\pi k}{N_s T}$$

3 Algorithm

In the following section, we derive the DPD algorithm for the location of OFDM signals. DPD is a centralised location method that is implemented at the network's mobile location center (MLC). The MLC is assumed to be the network function, which collects the received signals from all the BTS's in the network and estimates the emitter's fix. DPD is based on the extraction of the

$$\begin{aligned} \boldsymbol{r}_j(t) &= \boldsymbol{a}_j(\boldsymbol{p}) \cdot \frac{1}{\sqrt{T}} \sum_{\ell=0}^{\mathcal{L}-1} \alpha_j(\ell) \sum_{k=0}^{N_s-1} \bar{s}(k)\mathrm{e}^{j(2\pi/N_s T)k[t-\tau_j(\boldsymbol{p}_\ell)]} + \tilde{\boldsymbol{n}}_j(t) \\ &= \boldsymbol{a}_j(\boldsymbol{p}) \cdot \frac{1}{\sqrt{T}} \sum_{k=0}^{N_s-1} \bar{s}(k)\left(\sum_{\ell=0}^{\mathcal{L}-1} \alpha_j(\ell)\mathrm{e}^{-j(2\pi/N_s T)k\tau_j(\boldsymbol{p}_\ell)}\right)\mathrm{e}^{j(2\pi/N_s T)kt} + \tilde{\boldsymbol{n}}_j(t) \end{aligned} \tag{4}$$

$$\bar{\boldsymbol{r}}_j(k) = \boldsymbol{a}_j(\boldsymbol{p})\bar{s}(k)e^{-j(2\pi/T)k\tau_j(\boldsymbol{p})} \underbrace{\left(\sum_{\ell=0}^{\mathcal{L}-1} \alpha_j(\ell)e^{-j(2\pi/N_s T)k\bar{\tau}_j(\boldsymbol{p}_\ell)}\right)}_{\bar{b}_j(k)} + \tilde{\boldsymbol{n}}_j(k) \tag{6}$$

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position information from the signals that are simultaneously received at all the J BTS arrays.

The signals collected from the different BTS may or may not correspond to the same symbol. We distinguish between two BTS deployment configurations: 'BTS-sparse' and 'BTS-dense'. In a 'BTS-dense' network, the distance between the two most remote BTS arrays (with respect to each other) should be much smaller than the 'symbol propagation distance' [Note 1]. On the contrary, in a 'BTS-sparse' network, the distance between the BTS arrays is in the order of $T_s \cdot c$.

Thus, for a 'BTS-sparse' network, the assumption that the same symbol is received in all the BTS arrays simultaneously, is infeasible. Therefore in such a network, joint position estimation and symbol detection is impossible. However, localisation without joint detection is still possible.

Nevertheless, for most of the wireless systems, the distance between the intercepting BTS to every point within the coverage area is much smaller than the 'symbol propagation distance'. Hence, these systems may be treated as 'BTS-dense', for which we may assume that all the signals transferred to the MLC correspond to a specific symbol that was transmitted by the MSU. That enables us to apply the standard OFDM channel selectivity assumptions, which further allow the MLC to perform simultaneous signal detection and positioning. Clearly, that implies that channel estimation must be performed prior to emitter positioning. We shall now turn to the derivation of the DPD algorithm under the assumption of a BTS-dense network deployment.

Adhering to the common OFDM assumptions, we assume that there are L coherence bandwidths along the OFDM symbol where each coherence bandwidth applies to $\tilde{N} = \lfloor N_s/L \rfloor + 1$ consecutive subcarriers. It is further assumed that the OFDM symbol contains one pilot tone per coherence bandwidth. For simplicity, it is also assumed that on the uplink all the BTS observe the same coherence bandwidths. Although this assumption may not necessarily hold for all the cases, it is realistic for many suburban scenarios. Alternatively, one can assume that $B_C = \min\{B_{C_j}\}, j = 1, \ldots, J$, (i.e. apply the minimal B_C observed at the jth BTS to the other BTS as well). Following the 'piecewise-constant' assumption, we shall re-index the OFDM subcarriers using the indexes l and $\tilde{k}$ such that $k = \tilde{k} + l \cdot \tilde{N}$, $l = 0, \ldots, L-1$, $\tilde{k} = 0, \ldots, \tilde{N}-1$; that is, $\bar{s}(\tilde{k}, l)$, $\bar{\boldsymbol{r}}_j(\tilde{k}, l)$ and $\omega_{\tilde{k}l}$ correspond to the $\tilde{k}$th subcarrier in the lth coherence bandwidth. Given these definitions, the emitter's position $\boldsymbol{p}$ can be estimated by minimising the following cost function

$$Q(\boldsymbol{p}) = \sum_{j=1}^{J} \sum_{l=0}^{L-1} \sum_{\tilde{k}=0}^{\tilde{N}-1} \left\| \bar{\boldsymbol{r}}_j(\tilde{k}, l) - \boldsymbol{a}_j(\boldsymbol{p}) \bar{b}_j(l) \mathrm{e}^{-j\omega_{\tilde{k}l}\tau_j(\boldsymbol{p})} \bar{s}(\tilde{k}, l) \right\|^2 \tag{8}$$

Let $\tilde{b}_j(l)$ denote the complex channel attenuation coefficient of the lth pilot subcarrier at the jth BTS array. By re-casting the cost function in (8), the estimates of the complex channel attenuations associated with the lth pilot tones, $\tilde{b}_j(l)$, can be obtained as follows

$$Q_k(\boldsymbol{p}) = \sum_{j=1}^{J} \sum_{l=1}^{L} \left\| \bar{\boldsymbol{r}}_j(l) - \boldsymbol{a}_j(\boldsymbol{p}) \bar{s}(l) \mathrm{e}^{-j\omega_l \tau_j(\boldsymbol{p})} \tilde{b}_j(l) \right\|^2 \tag{9}$$

where the notations $\bar{s}(l)$, $\bar{\boldsymbol{r}}_j(l)$ and ω_l correspond to the pilot tone in the lth coherence bandwidth and are used here for brevity. It is further assumed that the exact index of that pilot tone is known to the MLC. The LSE of $\tilde{b}_j(l)$ is given by

$$\widehat{\tilde{b}}_j(l) = \left[\|\boldsymbol{a}_j(\boldsymbol{p})\|^2 \cdot |\bar{s}(l)|^2 \right]^{-1} \boldsymbol{a}_j^{\mathrm{H}}(\boldsymbol{p}) \bar{s}^*(l) \mathrm{e}^{j\omega_l \tau_j(\boldsymbol{p})} \bar{\boldsymbol{r}}_j(l) \tag{10}$$

Without the loss of generality, we assume that

$$\|\boldsymbol{a}_j(\boldsymbol{p})\|^2 = 1 \ \ \forall j, \quad |\bar{s}(l)|^2 = 1 \ \ \forall l \tag{11}$$

Thus, we obtain

$$\widehat{\tilde{b}}_j(l) = \boldsymbol{a}_j^{\mathrm{H}}(\boldsymbol{p}) \bar{s}^*(l) \mathrm{e}^{j\omega_1 \tau_j(\boldsymbol{p})} \bar{\boldsymbol{r}}_j(l) \tag{12}$$

Once the channel coefficients have been estimated, we can use them in order to obtain an estimate of the MSU location.

Let $\widehat{\tilde{\boldsymbol{b}}}_j$ denote the estimated vector of the complex channel coefficients at the jth BTS array. Under the piecewise-constant, block-selective fading assumption, we can define the following vectors and matrices

$$\widehat{\boldsymbol{b}}_j \triangleq \widehat{\tilde{\boldsymbol{b}}}_j \otimes \boldsymbol{1}_{\tilde{N}} \tag{13}$$

$$\boldsymbol{d}_j(\boldsymbol{p}) \triangleq [\mathrm{e}^{-j\omega_0 \tau_j(\boldsymbol{p})}, \mathrm{e}^{-j\omega_1 \tau_j(\boldsymbol{p})}, \ldots, \mathrm{e}^{-j\omega_{N_s-1} \tau_j(\boldsymbol{p})}]^{\mathrm{T}} \tag{14}$$

$$\widehat{\boldsymbol{B}}_j(\boldsymbol{p}) \triangleq \mathrm{diag}\left\{\widehat{\boldsymbol{b}}_j\right\} \cdot \mathrm{diag}\left\{\check{\boldsymbol{d}}_j(\boldsymbol{p})\right\} \tag{15}$$

$$\boldsymbol{A}_j(\boldsymbol{p}) \triangleq \widehat{\boldsymbol{B}}_j(\boldsymbol{p}) \otimes \boldsymbol{a}_j(\boldsymbol{p}) \tag{16}$$

where $\check{\boldsymbol{d}}_j$ corresponds to data subcarrier indexes of $\boldsymbol{d}_j$.

Define

$$\bar{\boldsymbol{r}}_j \triangleq \left[\bar{\boldsymbol{r}}_j^{\mathrm{T}}(0, 0), \bar{\boldsymbol{r}}_j^{\mathrm{T}}(1, 0), \ldots, \bar{\boldsymbol{r}}_j^{\mathrm{T}}(\tilde{N}-1, L-1) \right]^{\mathrm{T}} \tag{17}$$

$$\bar{\boldsymbol{s}} \triangleq [\bar{s}(0, 0), \bar{s}(1, 0), \ldots, \bar{s}(\tilde{N}-1, L-1)]^{\mathrm{T}} \tag{18}$$

Further, let $\check{\boldsymbol{s}}$ and $\check{\boldsymbol{r}}_j$ be vectors containing only the data subcarriers of $\bar{\boldsymbol{s}}$ and $\bar{\boldsymbol{r}}_j$, respectively.

Note 1: Symbol propagation distance $= 1.125 \times N_{\mathrm{FFT}}/f_s \cdot c = T_s \cdot c$, where $c = 3 \times 10^8$ m/sec is the signal propagation speed and f_s the OFDM signal sampling rate in Hz, N_{FFT} the FFT length; and the 1.125 factor accounts for a guard interval of 1/8 symbol. For example, for a sampling rate of 20 MHz and $N_{\mathrm{FFT}} = 64$, the symbol propagation distance is 1080 m.

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Next define

$$\check{\boldsymbol{r}} \triangleq \begin{bmatrix} \check{\boldsymbol{r}}_1 \\ \check{\boldsymbol{r}}_2 \\ \vdots \\ \check{\boldsymbol{r}}_J \end{bmatrix}, \quad \boldsymbol{A}(\boldsymbol{p}) \triangleq \begin{bmatrix} \boldsymbol{A}_1(\boldsymbol{p}) \\ \boldsymbol{A}_2(\boldsymbol{p}) \\ \vdots \\ \boldsymbol{A}_J(\boldsymbol{p}) \end{bmatrix} \tag{19}$$

The estimate of the emitter position is obtained by finding the vector $\boldsymbol{p}$ for which the following cost function is minimised

$$Q_u(\boldsymbol{p}) = \left\| \check{\boldsymbol{r}} - \boldsymbol{A}(\boldsymbol{p})\check{\boldsymbol{s}} \right\|^2 \tag{20}$$

The data subcarriers vector $\widehat{\check{\boldsymbol{s}}}$ that minimises the cost function (20) is given by

$$\begin{aligned} \widehat{\check{\boldsymbol{s}}} &= \boldsymbol{A}^{+}\check{\boldsymbol{r}} \\ &= \tilde{\boldsymbol{B}}\boldsymbol{A}^{H}(\boldsymbol{p})\check{\boldsymbol{r}} \end{aligned} \tag{21}$$

where $\tilde{\boldsymbol{B}}$ is given by (22).

By substituting (21) back into (20) and expanding the expression, the cost function can be recast as

$$\begin{aligned} Q_u(\boldsymbol{p}) &= \left(\check{\boldsymbol{r}} - \boldsymbol{A}\tilde{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}\check{\boldsymbol{r}}\right)^{\mathrm{H}}\left(\check{\boldsymbol{r}} - \boldsymbol{A}\tilde{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}\check{\boldsymbol{r}}\right) \\ &= \|\check{\boldsymbol{r}}\|^2 - \check{\boldsymbol{r}}^{\mathrm{H}}\boldsymbol{A}\tilde{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}\check{\boldsymbol{r}} - \check{\boldsymbol{r}}^{\mathrm{H}}\boldsymbol{A}\tilde{\boldsymbol{B}}^{\mathrm{H}}\boldsymbol{A}^{\mathrm{H}}\check{\boldsymbol{r}} \\ &\quad + \check{\boldsymbol{r}}^{\mathrm{H}}\boldsymbol{A}\tilde{\boldsymbol{B}}^{\mathrm{H}}\underbrace{\boldsymbol{A}^{\mathrm{H}}\boldsymbol{A}}_{\tilde{\boldsymbol{B}}^{-1}}\tilde{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}\check{\boldsymbol{r}} \\ &= \|\check{\boldsymbol{r}}\|^2 - \check{\boldsymbol{r}}^{\mathrm{H}}\boldsymbol{A}(\boldsymbol{p})\tilde{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}(\boldsymbol{p})\check{\boldsymbol{r}} \end{aligned} \tag{23}$$

Instead of finding the minimum of $Q_u(\boldsymbol{p})$, we can find the maximum of $\tilde{Q}_u(\boldsymbol{p})$ defined by

$$\tilde{Q}_u(\boldsymbol{p}) = \left\| \dot{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}(\boldsymbol{p})\check{\boldsymbol{r}} \right\|^2 \tag{24}$$

where

$$\dot{\boldsymbol{B}}^{\mathrm{H}}\dot{\boldsymbol{B}} = \tilde{\boldsymbol{B}} \tag{25}$$

The position $\boldsymbol{p}$ for which (24) is maximised, is the estimated location of the emitter. Thus, the final LSE for the emitter position, $\boldsymbol{p}$, is given by a 2D grid search

$$\widehat{\boldsymbol{p}} = \arg\max_{\boldsymbol{p}} \left\{ \tilde{Q}_u(\boldsymbol{p}) \right\} \tag{26}$$

Once the LSE for the position has been determined, the LSE for the transmitted OFDM symbol can be obtained by recalculating (21) for $\hat{\boldsymbol{p}}$. Further the detection of the modulating symbols is performed by a proper detector.

4 Discussion

Thus far, we have detailed the DPD method and outlined the traditional two-step AOA/TOA methods for estimating the position of an OFDM source. In both methods, the final emitter fix is done at an MLC of the network. However, DPD is a direct method in the sense that the MLC solves the localisation problem using the data collected at all sensors at all base stations together, whereas the traditional AOA/TOA approaches are composed of two separate steps: (1) AOA/TOA independent estimates and (2) triangulation based on the results of the first step.

The DPD estimator essentially correlates the expected array response of an emitter located at $\boldsymbol{p}$ with the observation. The expected array responses can be pre-measured, calculated in advance or calculated on-the-fly, for all the possible positions, $\boldsymbol{p}$, on a pre-defined search grid.

We notice that the DPD cost function, $\|\dot{\boldsymbol{B}}\boldsymbol{A}^{\mathrm{H}}(\boldsymbol{p})\check{\boldsymbol{r}}\|^2$, depends on three factors:

- $\boldsymbol{A}(\boldsymbol{p})$, a matrix containing the responses of all the arrays to an emitter located at position $\boldsymbol{p}$;
- $\check{\boldsymbol{r}}$, a vector containing the observed OFDM data subcarriers at all the arrays;
- $\dot{\boldsymbol{B}}$, a weighting matrix that depends on the channel conditions.

The grid point, $\hat{\boldsymbol{p}}$, for which the correlation between the observed signal and the expected array response is maximised at all the BTS simultaneously, is the least squares estimate of the emitter position.

Further noticing that the position vector $\boldsymbol{p}$ may consist of 2-D or 3-D coordinates implies that the extension of this algorithm from the 2-D to the 3-D case is straightforward.

$$\tilde{\boldsymbol{B}} \triangleq \begin{bmatrix} \left[\sum_{j=1}^{J}\left|\widehat{b}_j(0,0)\right|^2\right]^{-1} & 0 & \cdots & 0 \\ 0 & \left[\sum_{j=1}^{J}\left|\widehat{b}_j(1,0)\right|^2\right]^{-1} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \left[\sum_{j=1}^{J}\left|\widehat{b}_j(\tilde{N}-1, L-1)\right|^2\right]^{-1} \end{bmatrix} \tag{22}$$

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As will be shown in Section 6, the DPD outperforms the combined AOA + TOA estimation method (mainly under low signal-to-noise and interference ratio (SNIR) conditions). This result is not surprising as it is rather obvious that measuring AOA or TOA at each base station separately and independently is suboptimal since this approach ignores the constraint that all measurements must correspond to the same emitter position.

The advantages of the DPD method do not come for free: whereas AOA/TOA requires only the transmission of the measured AOA/TOA parameters to the MLC, the DPD technique requires the transmission of the raw samples of received emitter signals to the MLC. The requirement for raw data transmission obviously increases the network bandwidth consumed by the DPD compared with the traditional positioning methods. Although various data compression methods to reduce the DPD bandwidth consumption can be applied, the discussion of such methods exceeds the scope of this paper.

Despite the fact that here we have chosen to ignore the NLOS information, the DPD method can be modified to utilise the NLOS information. Since the NLOS response is summed up into the total array response, knowledge of the scatterers' location could be used in order to pre-generate replicas of the combined array response (i.e. LOS combined with NLOS). This combined array response could then be used in a similar fashion to improve the location accuracy. The incorporation of the NLOS information into the model requires channel model calibration, which implies that an accurate mapping of the entire location area is needed. In addition, it may require the application of geometric techniques such as ray tracing in order to accurately model the combined array response (including AOA and TOA of the multiple reflected rays that characterise the response pattern from each location on the defined search grid). It should be emphasised that the calibration method must produce a unique and unambiguous array response for each location on the search grid, otherwise, the DPD accuracy will degrade.

As a concluding remark, it should be noted that DPD can be applied to many types of signals including unknown signals. The choice of presenting a method for the OFDM emitter location was made because of the increasing popularity of OFDM modulation in the wireless community. The algorithm that has been developed takes advantage of the special structure of pilot and data tones, which constructs the OFDM symbol in order to enhance the performance of location estimation.

5 Paraboloid interpolation

The realisation of the LSE described in (26) is done by correlating the electrical field generated at the BTS array by the impinging emitter signal, with replicas of the electrical field, which would have been generated if the emitter were located at position (x, y), where the parameters x and y are defined over a grid with finite accuracy.

Practical location systems use pre-calibrated lookup tables for the estimated parameters, for example, AOA estimation systems use steering vectors for all the possible signal AOAs, and DPD-based systems hold lookup tables of the array response vectors. Such systems trade off the estimation accuracy, which is bounded by the grid granularity, for the search-time and memory requirements, which grow with the increase in the number of grid points. Thus, the reduction of the search complexity, while maintaining a high estimation accuracy, is highly desirable.

Enhancing the estimation accuracy with a finite grid can be achieved by performing interpolation across the extremum and its adjacent values detected in a grid-search procedure. In the world of one-dimensional (1-D) parameters estimation problems, the parabolic interpolation is a well known approach for obtaining higher estimation accuracy over a finite grid [21, 22]. Despite its similarity to the 1-D parameter estimation world, the field of fine parameter estimation in two-dimensional (2-D) problems has not received much attention in the literature.

The paraboloid interpolation, or bi-quadratic surface interpolation, which is the 2-D extension of the parabolic interpolation method is commonly used in image processing applications such as pattern recognition problems [23], or in systems involving motion estimation, where subpixel resolution processing is required [24, 25].

The paraboloid interpolation algorithm tries to fit a paraboloid surface to the virtual surface defined by the cost function peak $\tilde{Q}(\boldsymbol{p})$ detected in the 2-D coarse grid search and its eight surrounding cost-function values. Following the steps outlined by Gleason *et al.* in [24], the paraboloid surface is defined by

$$\tilde{Q}(\boldsymbol{p}) = a \cdot x^2 + b \cdot y^2 + c \cdot x \cdot y + d \cdot x + e \cdot y + f \tag{27}$$

The solution for the paraboloid coefficients' vector can be found by solving the system of equations

$$\boldsymbol{\Omega}\boldsymbol{\varphi} = \boldsymbol{\mu} \tag{28}$$

where

$$\boldsymbol{\Omega} = \begin{bmatrix} x_0^2 & y_0^2 & x_0 \cdot y_0 & x_0 & y_0 & 1 \\ x_1^2 & y_1^2 & x_1 \cdot y_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_8^2 & y_8^2 & x_8 \cdot y_8 & x_8 & y_8 & 1 \end{bmatrix} \tag{29}$$

$$\boldsymbol{\varphi} = \begin{bmatrix} a & b & c & d & e & f \end{bmatrix}^{\mathrm{T}} \tag{30}$$

$$\boldsymbol{\mu} = \left[\tilde{Q}(x_0, y_0), \tilde{Q}(x_1, y_1), \ldots, \tilde{Q}(x_8, y_8)\right]^{\mathrm{T}} \tag{31}$$

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Notice that indexes {0, . . . ,8} of the cost-function points in $\boldsymbol{\mu}$ must be consistent with the definition of $\boldsymbol{\Omega}$. We notice that the equation system defined in (28) is overdetermined and, hence, we apply the least-squares solution in order to extract $\boldsymbol{\varphi}$

$$\boldsymbol{\varphi} = \left(\boldsymbol{\Omega}^{\mathrm{T}}\boldsymbol{\Omega}\right)^{-1}\boldsymbol{\Omega}^{\mathrm{T}}\boldsymbol{\mu} \tag{32}$$

Once the paraboloid coefficients are known, the coordinates of the paraboloid maximum can be extracted by equating the gradients of $\tilde{Q}(\boldsymbol{p})$ with respect to x and y, to zero.

Another approach to the 2-D fine parameter estimation problem has been proposed by Giunta [26]. His algorithm utilises only five samples that surround the peak detected in the coarse grid search. It implies that the algorithm devised by Giunta ignores the additional information embedded in the three additional neighbouring samples, which are incorporated into the equation system of (28). As the numerical results presented in Section 6 will show, the oversight of the additional data results in biased estimates under high SNR conditions.

The application of paraboloid interpolation has two major advantages over the fine grid-search approach: it reduces the computational complexity of the fine search, (thereby, reducing the time required for locating the emitter), and it reduces the memory/storage requirements. The latter merit should also be viewed in the light of the calibration time requirement of a practical DPD system, which would decrease dramatically with the application of paraboloid interpolation.

We shall now demonstrate the reduced complexity advantage of using paraboloid interpolation over the fine grid-search approach by means of the following numerical example. First, we shall compare the computational complexities of the two approaches. The complexity of the paraboloid interpolation algorithm is fixed and is independent of the grid step size. To resolve the vector of paraboloid parameters, one has to calculate the product of two 9×6 matrices and use a straightforward matrix inversion, or Gaussian elimination in order to extract the vectors of the parameters. Since the Gaussian elimination requires about one-third of the operations compared with the matrix inversion approach, we shall consider it in the calculation.

It is easy to verify that the number of multiplications and additions required to calculate the product of two $(m \times n)$ matrices is $m \times n \times (n+1)/2$. The operation count for the Gaussian elimination procedure is given by Meyer [27, p. 10]: $2 \times n^3 + 3 \times n^2/2 - 7 \times n/6$. Thus, for the 9×6 matrix system of the paraboloid interpolation method, the total number of operations can be easily found to be 803.

In contrast, the number of multiplications and additions required by the DPD algorithm was approximated by Amar and Weiss [28]: $N_{\mathrm{p}} \times L \times M$, where N_{p} is the number of grid points. If we consider a fine grid of 11×11 cells, for a system that consists of four base stations, each equipped with an array of five antenna elements, then the total count of operations for the fine grid search would be 2420. Therefore in this example, the application of the paraboloid interpolation can reduce the computational complexity of the fine search step by nearly 70%.

The other benefit aforementioned is gained through the reduction in the system's memory requirements and calibration time. Memory requirements as well as calibration time tend to grow exponentially with every increase of magnitude in the grid resolution. Thus, the minimisation of these factors has a direct effect on the overall realisation cost of a DPD-based location system.

6 Numerical examples

To examine the performances of the proposed DPD algorithms and the accuracies of the various fine estimation approaches, we performed extensive Monte Carlo simulations. The location system consists of four BTS arrays placed at coordinates (−200, −200), (−200, 200), (200, −200) and (200, 200). All coordinates are in [metres]. Each BTS is equipped with a circular array of five antenna elements. The location of the LOS cluster containing the OFDM transmitter is selected at random, uniformly, within the square formed by the base stations. The NLOS interference is generated by additional three scattering clusters located at coordinates (−25, 0), (52, 15) and (95, 80). These locations are unknown to the MLC. The rays reflected from these clusters towards each of the BTS arrays, are attenuated (with respect to the LOS cluster) by 15, 25 and 35 dB respectively.

The transmitted signal is an OFDM signal that consists of 48 data tones and four pilot tones. Such a signal is typically used in IEEE 802.11a/g WLAN systems [29]. This case also represents subchannel allocation of IEEE802.16e WiMAX systems [12]. Each OFDM tone is modulated with a narrowband quadrature phase shift keying (QPSK) complex symbol. The modulating symbols for the data tones are selected at random and are unknown to the MLC, whereas the pilot tone modulating values are provided to the MLC.

The complex attenuations for the pilot tones are selected at random using complex-normal distribution (mean = 1, std = 1) and assigned to the $\tilde{N}$ neighbouring data tones as well. Different attenuation vectors are generated for each of the J BTS.

Each location determination is based on a single observation of the OFDM subcarriers. In addition to the NLOS interference, the received symbol is also subjected to complex, additive, white-Gaussian noise. The generated noise plus interference is added to the attenuated signal on

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every frequency bin and each antenna element at each of the J antenna arrays.

The simulation compared the traditional AOA + TOA location method with the proposed DPD algorithms. The AOA estimates are obtained as described in [30]. A description of the TOA estimation can be found in [31] and [1], and that of the cost-function construction for the combined AOA and TOA positioning is outlined in [32].

In Figs. 1–4, the position determination is done via a grid search, where the search is performed in two steps: first, a coarse search with a 10 m resolution over the full area of interest, which is followed by a fine estimation of the position. The final position estimation of the DPD compares the three approaches outlined in this paper, namely: a fine search with a 1 m resolution near the peak found in the coarse search, a fine estimation with the paraboloid interpolation and a fine estimation according to the method devised by Giunta, which is outlined in [26].

At each SNIR value, we performed 500 experiments in order to obtain the statistical properties of the performance. The performance evaluation is based on the statistics of the miss distance, that is, the distance between the true emitter position and the estimated emitter position defined by

$$\text{Error} = \|\hat{\boldsymbol{p}} - \boldsymbol{p}\| = \sqrt{(\hat{x} - x)^2 + (\hat{y} - y)^2}$$

We used two different criteria:

1. The Root mean square of the miss distance (Figs. 1 and 3).

2. The mean of the miss distance (Figs. 2 and 4).

Although the assumption of LOS with four BTS holds in some networks such as WLAN networks that can be considered as 'access-point dense', it may not hold for outdoor cellular networks. Therefore we have also evaluated the location performance with only three BTS arrays. The results are depicted in Figs. 3 and 4. In this simulation scenario the BTS array at (200, 200) has been eliminated and the location was estimated using the signals received at the three remaining BTS arrays.

In addition, the detection performance of the proposed algorithm are given in Section 5, which depicts the QPSK symbol error rate (SER) performance for the detected OFDM symbol. The results presented in Fig. 5 were obtained for a three-BTS location system under the NLOS channel conditions aforementioned and for an OFDM symbol consisting of four pilot tones. As can be seen, the SER performance under low SNIR conditions are poor.

Several conclusions can be evidently drawn from the figures. First, it is clear that the proposed DPD method outperforms the traditional AOA + TOA approach. The main advantage is observed under low SNIR conditions. Such conditions typically correspond to a dominant NLOS

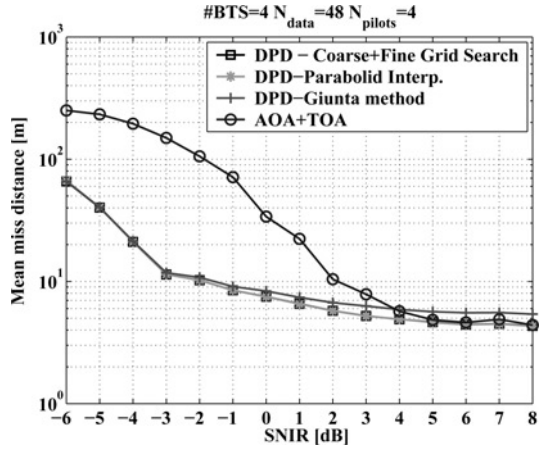

Figure 2 *Mean miss distance for DPD/AOA + TOA operated on an OFDM signal with four BTS arrays*

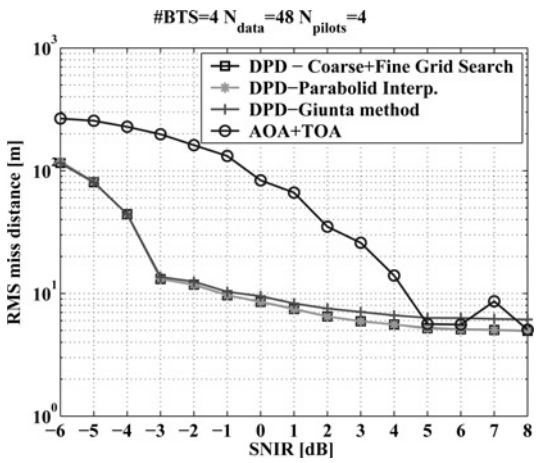

Figure 1 *RMS miss distance for DPD/AOA + TOA operated on an OFDM signal with four BTS arrays*

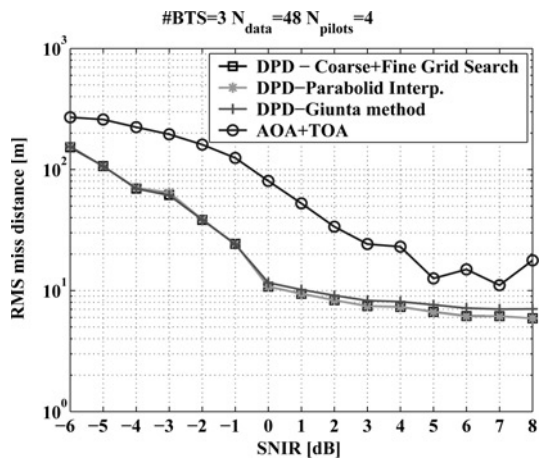

Figure 3 *RMS miss distance for DPD/AOA + TOA operated on an OFDM signal with three BTS arrays*

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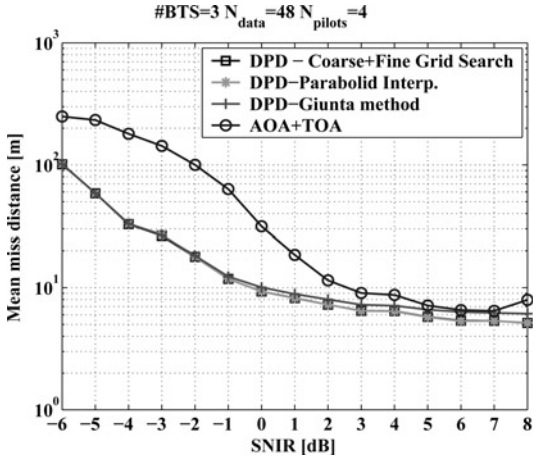

Figure 4 *Mean miss distance for DPD/AOA + TOA operated on an OFDM signal with three BTS arrays*

interference scenario. Secondly, it can be seen that the paraboloid interpolation approach for fine estimation yields a nearly identical statistical performance to the coarse + fine grid search.

Thirdly, both the DPD and AOA + TOA methods exhibit irreducible error under high SNIR values. This can be attributed to the NLOS interference, which remains regardless of the noise power and, hence, results in an 'error floor'.

Fourthly, as it can be seen, for example, by comparing the results in Figs. 1 and 3, the elimination of the fourth BTS is manifested in a performance loss of about 3 dB. This result is expected because of the loss in diversity and is true for both the DPD and the AOA + TOA methods.

Lastly, whereas the fine estimation approach devised by Giunta results in biased estimates under high SNIR, under a low-SNIR regime, this method yields a nearly similar performance to the other two fine estimation methods. Considering its lower computational complexity (as it does not require any matrix inversion that is required by the paraboloid interpolation method, but only some additions and a couple of divisions), the Giunta method can be considered as an adequate choice for fine estimation in location systems that are designed to operate under a low-SNIR regime. Nevertheless, its estimations under higher SNIR are biased, and it is, thus, inferior to the paraboloid interpolation method.

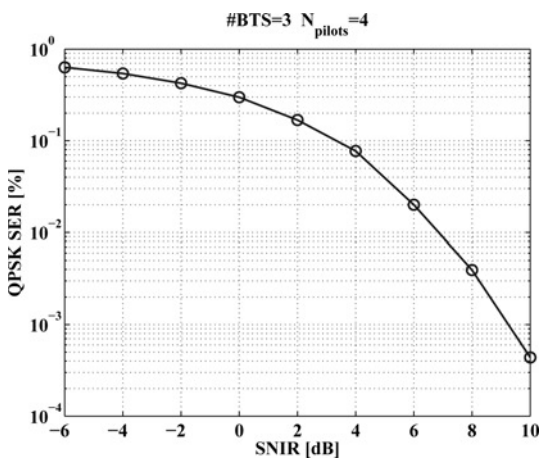

Figure 5 *OFDM data subcarriers SER with three BTS arrays*

7 Conclusions

An efficient, least-squares algorithm for DPD of the OFDM signals has been presented. It has been shown that the proposed algorithm outperforms traditional location techniques such as AOA + TOA. The DPD algorithm is realised by using a 2-D grid-search process. The initial position estimate is obtained from a search over a relatively coarse grid. Three techniques for refining the estimation performance have been discussed: coarse+fine grid search, paraboloid interpolation and the 2-D fine peak estimation method devised by Giunta [26]. It has been shown that under high SNIR, the paraboloid interpolation algorithm outperforms the Giunta-proposed 2-D fine estimation algorithm and achieves an estimation performance that is similar to those that can be achieved by employing a second fine grid-search step. The advantage of the paraboloid interpolation method is that, because of its reduced memory requirements, it enables the realisation of a much simpler and cheaper location system. Nevertheless, for systems that are targeted for a lower SNIR regime, the Giunta method can be considered as a suitable choice as it offers lower computational complexity compared with the paraboloid interpolation, with nearly similar statistical performance.

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Amendment 4: Further Higher Data Rate Extension in the 2.4 GHz Band

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